

## # Boston Housing Price Prediction

This project demonstrates a machine learning pipeline for predicting housing prices in Boston using the Boston Housing Dataset. It includes both model development and a user interface for interaction.

### ## Contents

- `BostonHousing.ipynb`: Builds and trains a regression model using Scikit-learn.
- `BostonInterface.ipynb`: Provides a graphical user interface (GUI) for making predictions using the trained model.

---

### ## 1. `BostonHousing.ipynb`

#### ### Description

This notebook covers the complete workflow for predicting housing prices:

- Loading the Boston Housing Dataset
- Data preprocessing and exploration
- Feature selection and transformation
- Model training using Linear Regression
- Evaluation of model performance ( $R^2$ , MSE, etc.)
- Saving the trained model for later use (`joblib`)

#### ### Libraries Used

- `pandas`

- `numpy`
- `seaborn`, `matplotlib`
- `sklearn` (for model building and evaluation)
- `joblib` (to save the model)

---

## ## 2. `BostonInterface.ipynb`

### ### Description

This notebook creates a basic interface using `ipywidgets` for user interaction:

- Loads the saved model from the training notebook
- Allows the user to input feature values through sliders or text fields
- Displays the predicted housing price based on user inputs

### ### Libraries Used

- `ipywidgets`
- `IPython.display`
- `joblib`
- `sklearn`
- `numpy`

---

## ## How to Run

### 1. **\*\*Install Dependencies\*\***:

```
```bash
```

```
pip install numpy pandas matplotlib seaborn scikit-learn ipywidgets joblib
```

```
```
```

2. **\*\*Run `BostonHousing.ipynb`\*\*:**

- Trains the model
- Saves it as `boston\_model.pkl`

3. **\*\*Run `BostonInterface.ipynb`\*\*:**

- Loads the saved model
- Allows you to input values and get a predicted price

```
---
```

## ## Requirements

- Python 3.7+
- Jupyter Notebook
- All required libraries as listed above

```
---
```

## ## License

This project is open-source and free to use for educational purposes.